Seeing more morphology: Are form-meaning relations in British Sign Language grammar iconic for hearing non-signers?

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Iconicity is a core property of both spoken and signed languages (Perniss et al., 2010) but it has been claimed that it is particularly responsible for driving the emergence and evolution of grammatical structure in sign languages (Aronoff et al., 2005; Sandler et al. 2011). Beyond this role in emerging sign languages, many established signing communities have a large proportion of signers with variable sign language acquisition experiences, and it has been suggested that this may have an impact on sign language grammatical structure (e.g., Schembri et al., 2018). For example, as suggested by Trudgill (2011), the relatively large proportion of signers who acquire the language in late childhood or early adulthood may result in a greater proportion of systematically iconic and highly learnable structures. In this presentation, as a first step towards investigating this claim, we explore novel ways to answer questions about the nature and role of iconicity in sign language grammar, with a focus on British Sign Language (BSL).

In the study presented here, hearing non-signers' sensitivity to the iconicity of several different morphological modifications in BSL are tested, aiming to contribute further evidence about whether iconicity in sign language morphology arises due to aspects of shared human cognition. The BSL morphological patterns that serve as the focus of this study include verb directionality (modification of the base verb's movement path to show subject and object arguments), verbal plurality (signalled by modifications of the base verb sign's movement), aspect marking (using fast and slow reduplication of the base verb), and non-manual modification (puffed cheeks vs. tongue protrusion co-occurring with the base verb

or adjective sign to signal different adverbial meanings). We believe that this study represents the first time these aspects of sign language morphology have been investigated in this way.

In a previous study, we found that the meaning of many BSL morphological modifications appears to be guessable in a forced choice task even without language knowledge. However, we queried whether that original task had artificially high rates of accurate responses because some or all of the distractor items were too different from the target answer, thus more easily guessable. As a result, the task was modified and improved by including distractor items based on responses provided by participants in an open response version of the same task. For each item, the distractor set consisted of two options assessed as "part correct" and one assessed as "incorrect". The final forced choice task presented here contained a more consistent set of semantically related competing distractors alongside the target response.

100 hearing British non-signing participants were recruited for this version of the study. For each of 40 items, participants were presented with a citation form video accompanied by an English equivalent and were then asked to pick one of four possible English translations to match a second video of the sign with a morphological modification. 10 participants were excluded due to failing an attention check, reporting technical issues, or reporting knowledge of a sign language. Data from 90 participants (42 female, mean age 46 years) was analysed (see Figure 1).

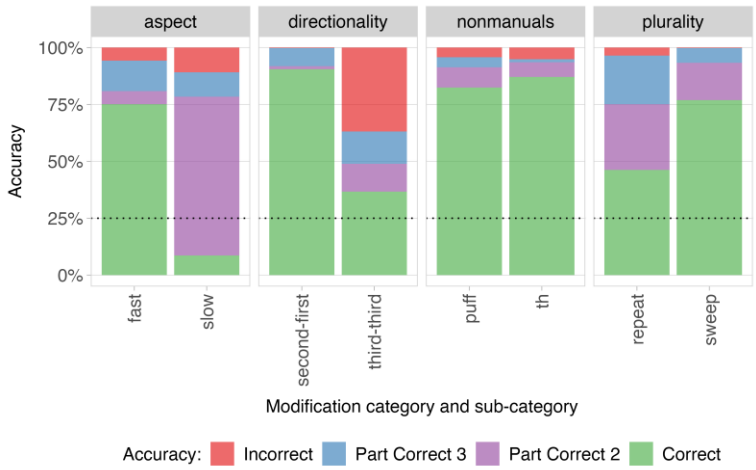

Figure 1 Response Accuracy Split by Modification Category and Sub-Category.
Note: Horizontal dotted line shows chance level of guessing the correct answer from the four multiple choice options.

A Bayesian binomial analysis of the results suggests that not all types of iconicity are equally transparent. Both of the non-manual modifications were easily guessed, but the aspect marking, directionality, and plurality results were mixed. While the meaning of the second-first person directionality modification was easily guessable (a reversal of the demonstrated first-second direction citation form), the third-third person modification results were close to chance with around one third of participants identifying the modified sign's directionality as second-first - the correct answer for the second-first directionality modification. The meaning of the fast aspect reduplication (to signal a repeated and/or habitual action) was relatively transparent, but the results for the slow movement (intermittent action) modification were below chance level. Here again, the most frequently selected option matched the correct answer for fast aspect reduplication. For plurality modifications, repeat was harder to guess than sweep; while the contrast was less pronounced than for aspect and directionality, the difference was still statistically credible. Overall, we observed a moderate but credible learning effect.

Together, these findings suggest that iconicity in sign language morphology is partly accessible without sign language knowledge due to shared aspects of human communicative behaviour, such as the use of pointing gestures and signs directed towards the body to represent first person participants. Showing that hearing non-signers can access iconicity in morphological structures suggests it may be a core property in language emergence and evolution. However, in methodological terms, the iterative design of this study also reinforces the point that the semantic range of the distractors and targets in closed response tasks is a crucial factor.

References

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